From Embodied Experience to Persistent Experience-Derived State: An Auditable Unified-Graph Architecture and a Preregistered Boundary Analysis of Activation-Based Access

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ABSTRACT

Embodied agents accumulate experience far faster than they can write it back as persistent state: action histories remain logs. We present FASCINATOR (FAS), a cognitive-architecture prototype that keeps semantic, episodic, and procedural content on a single typed, weighted graph and writes experience back as relational structure through one evidence-weighted attribution pipeline. The paper contributes an auditable memory architecture together with a preregistered evaluation protocol that delimits when spreading activation—the architecture's access mechanism—adds value over flat retrieval. In an information-asymmetric beacon task, persistent graph context yields 20/20 task completions versus 0/20 without context (McNemar $p{=}2\times10^{-6}$; the 0/20 is a closed-book design's built-in existence verification); an activation-free embedding-retrieval control reproduces the full benefit (20/20). Persistent experience-derived state is therefore supported as behaviorally effective, while the necessity of activation is not. A context-removal control experiment establishes that persistence itself is causally operative: after graph reset, goal resumption drops from 10/10 to 0/10 (exact McNemar $p{=}0.00195$, the minimum attainable at $n{=}10$) and repeated failures on an invalidated goal rise from a mean of 2.8 to 7.7 (one-sided Wilcoxon $p{=}0.00098$); whenever the experience stream remains re-readable, history and retrieval baselines match FAS on goal resumption and exceed it on failure repetition (0.0 vs. 2.8 repeats; Section 5.6)—so the causal role attaches to persistence itself, while the independent contribution of graph structure and activation remains open. A preregistered mechanism campaign (4×50 paired seeds) rejects activation's incremental value over a closed-form flat baseline (ΔJCG<0, Holm $p{=}1.5\times10^{-9}$, 0/200 reversals), isolating one exception: multi-input convergence through a shared hub (FAS 30/30 *vs* flat 0/30). An 84-cell parameter sweep maps this exception—FAS strictly beats flat in 11 cells (weak edge weights, shallow paths), loses in 9 (strong weights, deeper paths), and ties in 64. A second, baseline-controlled campaign (508 archived runs; 390 repair runs archived separately) measures the trade-off directly: under neutral instructions in the short-history regime, the persistent state resumes an abandoned goal in 9/10 runs where full-history LLM baselines do so in 0/10 (longer, failure-containing streams: baselines catch up at 10/10), and the same persistence repeats an invalidated goal 10.5 times on average where the baseline pivots after 1.0 (a different, unwired assembly; Sections 5.5–5.6). All negative results, two assembly audits, and the full protocol are reported.

1 INTRODUCTION

As soon as an embodied agent acts for more than a few minutes, it produces a stream of experience: actions, observations, successes, failures. What happens to that stream separates two classes of system. In

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one class, experience accumulates as logs—available for reflection prompts or retrieval augmentation, but never re-organized into structure that changes future behavior. In the other, experience is consolidated: concrete episodes give way to general relational knowledge that persists, is re-activated in relevant situations, and measurably participates in later decisions. Whether or not the system's core includes a large language model (LLM), the gap between logs and consolidation is a central engineering problem for long-lived agents.

Existing positions are familiar. Cognitive architectures such as ACT-R (Anderson et al., 2004) and Soar (Laird, 2012) consolidate experience through specific mechanisms (chunks, reinforcement) but keep memory stores representationally separate. LLM-based agents (Yao et al., 2023; Park et al., 2023; Shinn et al., 2023) typically save experience as text or vectors and reuse it through prompting; consolidation is delegated to the model and is not auditable. A graph offers a middle path: if episodic, semantic, and procedural content share one explicitly typed graph, the write-back of experience—two named layers, continuous Hebbian reinforcement and thresholded structural promotion (terminology note in Section 3.3)—becomes a first-class, auditable operation rather than a side effect of prompting.

This paper reports FASCINATOR (FAS), a Python prototype built around that idea: a *unified cognitive graph* on which spreading activation (Collins and Loftus, 1975) drives attention and candidate generation, an attribution pipeline writes experience back as typed relational structure, and ledger and event-framework channels route stored knowledge into behavior. We deliberately keep two propositions separate throughout:

(a) **Persistent relational reuse has behavioral value.** Under information asymmetry—the knowledge needed for a decision exists only in memory, not in the current observation—graph context built from experience changes behavior (beacon task: 20/20 completions vs 0/20 without context).

(b) **The necessity of activation is not supported.** The same benefit is fully reproduced by activation-free retrieval (D-flat, 20/20), and a preregistered mechanism campaign rejects activation's incremental value over flat retrieval under all four confirmatory conditions.

The contributions are:

1. **An auditable unified-graph memory architecture.** Semantic, episodic, and procedural content coexist in one typed weighted graph; experience becomes new relational structure only through a single evidence-weighted write-back channel. Isolation experiments verify the channel (0 vs 13 learned nodes under an identical experience stream), cross-context re-activation (4/12 vs 0/8 vs 0/0 marked nodes), and two consumer channels that measurably flip action selection (Sections 3–4.2).

2. **A preregistered evaluation protocol with honestly reported negative results.** The campaign testing whether spreading activation adds value over flat retrieval was designed, seeded, and committed before execution; its primary hypothesis was rejected (ΔJCG<0 under all four preregistered conditions, Holm $p=1.5\times10^{-9}$, 0/200 reversals), and we report this as a primary result (Section 4.3).

3. **A boundary analysis of activation-based access.** The single systematic exception—multi-input convergence through a shared hub (30/30 vs 0/30)—is delimited by an 84-cell parameter sweep over edge weight, input count, and path depth, and complemented by a structural diagnosis of relation-direction semantics (Section 4.3, 5).

Every number in this paper is traceable to archived run outputs or to the output of the analysis scripts that read them: the headline numbers are regenerated by the one-command verifier shipped with the reproducibility package, and the remainder were checked line by line against the archives (audit trail in the package and in the reproducibility report); the evidentiary standard, the deterministic-sandbox protocol, and the statistics are specified in Section 3.7. Internal codenames, per-family provenance, and full per-experiment data are in the Supplementary Material.

2 RELATED WORK

Cognitive architectures. ACT-R (Anderson et al., 2004) and Soar (Laird, 2012) treat consolidation as mechanism-specific learning over separate memory stores; OpenCog (Hart and Goertzel, 2008) shares our atom-space motivation but targets general intelligence rather than auditable consolidation of embodied

experience. Episodic-memory theory (Tulving, 1983; Baddeley, 2000) and event segmentation (Zacks et al., 2007; Schacter et al., 2007) motivate our event-framework channel and attribution window. Spreading activation as an access mechanism goes back to Collins and Loftus (1975); attention and working-set selection relate to Itti and Koch (2001); Posner (1980); Cowan (2001).

LLM-agent memory. ReAct (Yao et al., 2023), Generative Agents (Park et al., 2023), Reflexion (Shinn et al., 2023), and RAG (Lewis et al., 2020) reuse experience through prompting or vector retrieval; consolidation quality is not separately controllable or auditable. Structured-memory systems come closer: HippoRAG (Gutiérrez et al., 2024) and GraphRAG (Edge et al., 2024) organize retrieval over graph-structured indices but do not test activation-based access against flat retrieval on the same structure; MemGPT (Packer et al., 2023) manages tiered memory through function calls rather than a persistent relational graph; Voyager (Wang et al., 2023) stores reusable Minecraft skills as generated code, an explicit skill library rather than an auditable graph write-back. FAS uses the LLM only as a budgeted interface for language and never as memory (Section 3.5); the studied knowledge channels run with the LLM hard-disabled.

Relation to our contribution. To our knowledge, none of these lines reports a preregistered test of whether activation-based access beats flat retrieval on the same structure. Our negative result on that question, and the parameter sweep delimiting the exception, are the paper's methodological core.

3 MATERIALS AND METHODS

We present the system content-first: the graph, activation, write-back, persistence and consumers, then the LLM boundary.

3.1 The unified cognitive graph

FAS maintains a directed, typed, weighted graph $G = (V, E, W)$. Nodes carry a label τ_v marking memory role (seven classes—items, recipes, action experiences, changes, events, intentions, gaps—sharing one namespace), a base weight w_v, a timestamp, and a runtime activation $a_v \in \mathbb{R}$; each edge $e = (u, v, \rho, w)$ carries a relation type ρ (eight classes: produces, belongs-to, drops, needs, needs-tool, leads-to, involves, results) and a static weight $w \in [-2, 2]$; negative weights denote suppression. *There is no separate episodic, skill, or intention store*: goals, events, action patterns, and affect are nodes of the same graph, distinguished only by label and relation type (Fig. 1). Beside the graph there are only two auxiliary layers—a short-term episodic buffer (capacity 100, for short-horizon integration) and a statistical timeline (the ledger)—which never build structure: the sole structure-writing channel is the attribution pipeline of Section 3.3. The space split serves lifecycle management only; it does not isolate topology.

3.2 Cognitive state and spreading activation

The cognitive state at time t is the activation field $\mathbf{a}(t)$. Inputs (observations, intentions, gap injections) deposit initial energy into the corresponding nodes; each step, every active node u emits to its neighbors

$$\Delta a_{u\to v}(t) \;=\; a_u(t) \cdot \tilde{w}_{uv} \cdot \beta_{\text{cat}}(\rho) \cdot g_{\text{rel}}(\rho) \,\Big/\, \textstyle\sum_{v'} \max\big(\tilde{w}_{uv'}, 0\big), \tag{1}$$

followed by decay $a_v(t+1) = (1-\lambda)\, a_v(t) + \sum_u \Delta a_{u\to v}(t)$, clamped by per-space maxima (episodic space $a_{\max}$=3, global default 5). Emission follows an "emit = transfer" policy: each node's emitted quantity per step (default 1.0) is deducted from itself; source nodes are exempt. This is an engineering choice of short memory, not an energy-conservation law. Negative-weight edges contribute suppressive flow; suppression is issued only when the emitting node's positive-weight total is greater than zero. The emission budget is scaled by a modulatory gain

$$g \;=\; \text{clip}\big(1 + 0.3\,\text{arousal} - 0.15\,\text{stress},\; 0.7,\; 1.6\big), \tag{2}$$

where arousal is derived from affective-node activation and stress from a slow tonic offset. Attention is the Top-k of the field and constitutes input to action selection. Fig. 2 sketches activation-based access *side by side with its flat-retrieval comparison*; the schematic builds intuition only—all conclusions are carried by the experiments of Section 4.3.

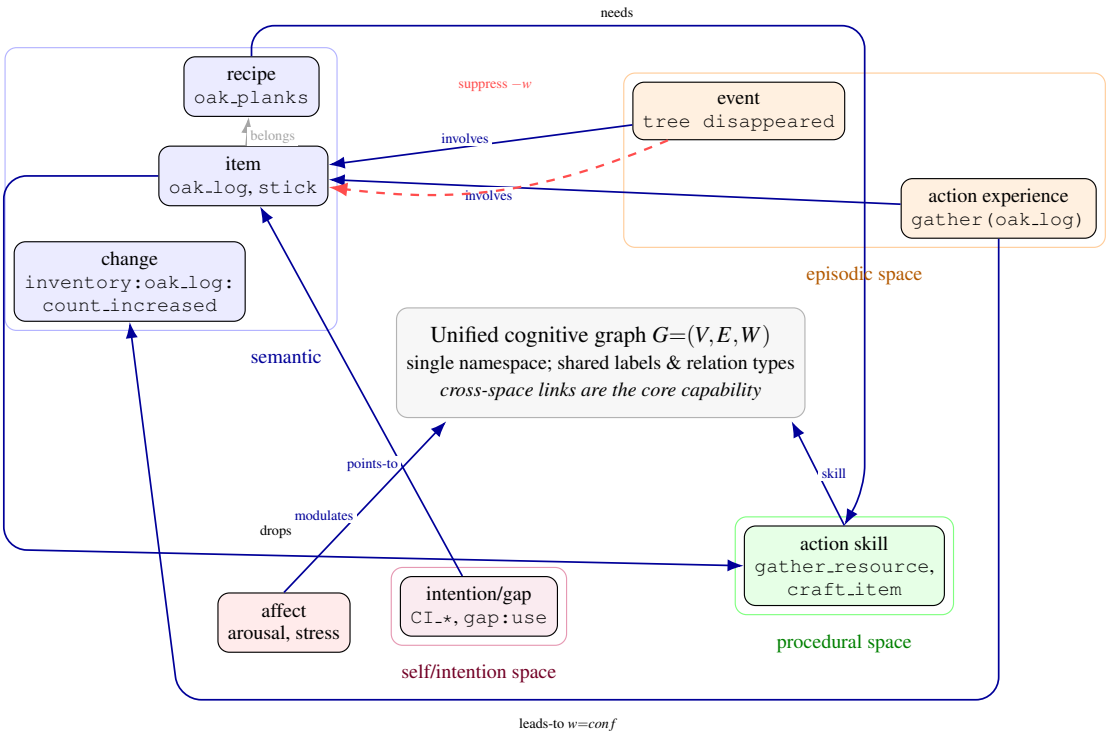


Figure 1. The unified graph ontology (schematic). Seven label classes and eight relation types share one namespace; there is no separate episodic, skill, or intention store. Cross-space links (blue, corridor-routed) are the core capability: the `leads-to` edge from `action experience` to `change` (weight = confidence) is the product of the write-back pipeline (Section 3.3); `gap` points to an entity to inject activation; `event` carries a negative-polarity slot edge (red dashed) suppressing the entity (Section 3.4).

3.3 Experience attribution and write-back

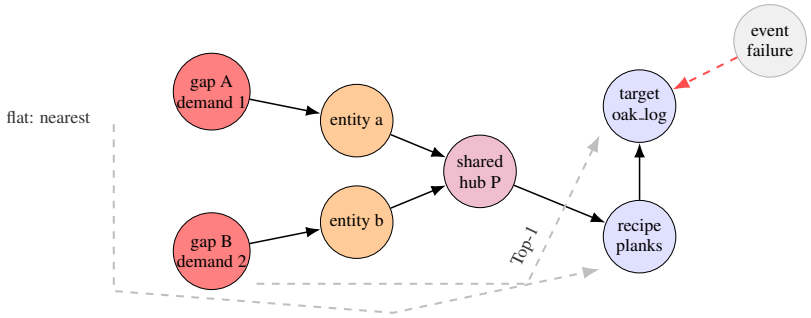
After an action settles, its receipt and the attribution window (8–120 s of environment time) of observations form candidate action–observation pairs. Repeated pairs aggregate into hypotheses carrying support s, contra-evidence c, and Laplace confidence $\text{conf} = s/(s+c+2)$. When $s \geq 5$ and $\text{conf} \geq 0.80$, the hypothesis is *promoted* to a typed relation on the graph: a `leads-to` edge of weight $w = \text{conf}$ from the action node `action` $:A(x)$ to the change node `change` $:O$,

$$\texttt{action}:A(x) \longrightarrow \texttt{change}:O \qquad (\texttt{leads-to},\ w{=}\text{conf}), \tag{3}$$

and both endpoints are mark-activated. This is how experience becomes *new relational structure*: `action:`/`change:` nodes exist only as products of this pipeline. *Terminology note.* This paper distinguishes three operations: *continuous reinforcement* (sub-threshold Hebbian co-activation weight adjustment; mechanism measurements in Sections 4.6 and 4.7), *episodic replay consolidation* (reorganization driven by offline replay, implemented in the intention/replay layer but not behaviorally validated here), and *structural promotion* (the thresholded hypothesis → graph write defined in this section). The blanket term "consolidation" used in the introduction and related work names the problem-level ideal only; every claim about FAS below is made under the name—and within the evidence scope—of one of these three operations. The same ledger maintains per-action success-rate memories, consumed by action selection as prior bonuses (± 0.05)—the second statistical path from experience to behavior. A third channel encodes discrete events as event-framework nodes with polarity-slot edges (implemented and production-default off; this paper assembles it explicitly, Section 4.2). The algorithm is Algorithm 1.

3.4 Persistence and consumers

Knowledge survives episode boundaries by construction (the graph is inherited, not reinitialized); persistence carries two payloads: *structure* and *activation values* are both serialized into the next episode, so "re-activation across contexts" is a measurable event. Section 4.2 measures that re-activated ($a \geq 0.05$)



Diffusion paths: multi-source input converges via relational structure on hub P—downstream convergence (Section 4.3):
- mechanism runs: JCG>0 on 200/200 paired seeds; ≥2-input MRR@1=1.0 (hypothesis group)
- only systematic positive difference: shared-hub topology, FAS 30/30 *vs* flat 0/30 (single weight w=0.5 design)

Flat path: nearest-neighbor retrieval reaches the target directly—sufficient for ordinary retrieval:
- ΔJCG<0 under all 4 preregistered conditions (reverse-direction Holm $p=1.5\times10^{-9}$, 0/200 reversals)
- wider spread dilutes targets ($\rho=-0.445$, -0.656); *not drawn as diffusion-always-wins*.

Figure 2. Spreading activation and flat retrieval side by side. Every numbered claim comes from the archived experiments of Section 4.3; the figure only builds intuition.

Algorithm 1 Attribution–write-back: experience to relational knowledge

Require: action–observation stream (a_t, o_t) within the window; ledger $\mathscr{L}$; graph G

For each action A_t, associate observations O_k in the window with its settlement as candidate pairs (A_t, O_k)
Aggregate by action signature: $s \mathrel{+}= 1$ (positive outcome), $c \mathrel{+}= 1$ (contra)
conf $= s/(s+c+2)$ (Laplace smoothing)
if $s \geq 5$ and conf ≥ 0.80 **then**
    create `action`:A, `change`:O nodes if absent
    write edge `action` $\longrightarrow$ `change` with relation `leads-to`, weight w=conf; mark both active
end if
ledger in parallel maintains per-action success priors $\mathscr{L}$.prior(sig(A)) consumed by action selection

differences are carried exactly by episodic structure connecting shared entities (4/12 vs 0/8 vs 0/0). Snapshot rotation and a shrink guard (save refusal to overwrite a running graph with an empty one) are data-integrity mechanisms, not cognitive content.

Four classes of consumer use the knowledge produced by write-back. **(a) Ledger channel (production):** candidate action scores are reweighted by ledger success priors (+0.05 success / −0.05 failure). **(b) Affect/neuromodulation (production, every tick):** arousal and stress scalars modulate the emission gain via Eq. 2. **(c) Continuous intentions and replay gating (production):** the continuous-cognition cycle runs graph dynamics every 2.5 s; intention lifecycle and idle-gated offline experience replay belong here. **(d) Event framework (experimental assembly, production-default off):** negative events suppress entity activation via polarity-slot edges; the code is production-hardened but disabled by default so that nothing lands on the graph in production; all behavioral evidence comes from experimental assembly. *Experimental layers* (sandbox assembly and the LLM decision head of Section 3.7.4) never enter the production default path; the run recorder only observes—it never modifies cognitive behavior.

3.5 The LLM as a budgeted interface

FAS uses the LLM for language production, interpretation of unstructured input, and (low-frequency) reflection; it never uses the LLM as memory, nor as the source of the knowledge this paper studies. All subject experiments run under an import-time guard that hard-disables the LLM (zero-LLM contract; any call errors out)—the only exceptions are the explicitly labeled decision-head experiments (Section 3.7.4).

3.6 The experience-to-knowledge loop

Fig. 3 shows the complete conceptual loop. Experience is segmented into temporally ordered action–observation fragments (the attribution window), repeated patterns accumulate evidence into candidate relations, and, after support and confidence checks, are written back as persistent graph structure. New

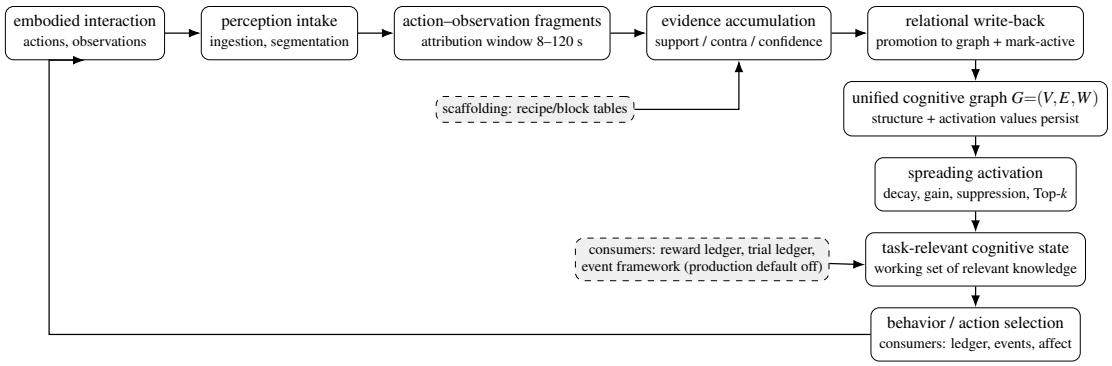


Figure 3. The FAS conceptual loop: from embodied interaction to reusable relational knowledge, and back to behavior via activation and consumers. Dashed blocks are scaffolding (mechanics tables; the necessary precondition characterized by the zero-prior result, Section 4.7) and consumption channels. The LLM's only positions in the loop are language production/interpretation/reflection (Section 3.5).

inputs unfold across this graph via spreading activation, forming a task-relevant cognitive state with a Top-k focus; consumers carry that state into action selection; actions change the environment, producing new experience. Language production, interpretation, and reflection are the only LLM calls in the loop (Section 3.5): the LLM is a consumer/decision component, not memory.

3.7 Evaluation protocol

3.7.1 Environment, determinism, and what "paired seeds" mean

All experiments run in a deterministic `SandboxWorld`: a discrete Minecraft-style grid world with a simulated clock (16 s/tick), recipe and block tables, and a fake device bridge replacing a real Minecraft LAN bridge. The agent stack is the production stack—knowledge graph, diffusion engine, continuous-cognition components, autonomy loop, action manager, and the real skill layer—with the LLM disabled by the import-time guard. One long-chain condition uses a documented simplification (sandbox smelting settles instantly into the inventory); results relying on it are marked.

Every condition is instantiated with a fixed seed; for the zero-LLM subject experiments, identical seeds reproduce bit-identical trajectories (verified by re-runs). For the LLM decision-head families, see the determinism re-run in Section 4.4. *Paired seeds* instantiate different random instances of the same condition and support paired statistics across instances; because the world is a deterministic sandbox with synthetic topologies, they do *not* sample real-world variance. Statistical tests in this paper therefore quantify the consistency of a direction of difference across instances, not population-level effect sizes. Sample sizes: subject experiments n=5 seeds; mechanism campaign n=50 paired seeds per condition; beacon and routing n=20 paired seeds; relation-direction diagnosis n=10 paired seeds. Every run is archived by the run recorder (metadata, event stream, start/end graph snapshots, metrics) into content-addressed directories; the archival census is given in the Supplementary Material (Table S2).

3.7.2 Preregistration and evidence accounting

The mechanism campaign's hypotheses, topologies, seed reservoir, metrics, and testing scheme were fixed and committed before execution; deviations (one assembly repair before unblinding) are reported as such. Evidence is reported at three levels—*representational* (existence and re-activation of learned nodes/edges), *mechanism* (activation trajectories, weight changes, ledger counts), and *behavioral* (completion ticks, action counts, first-action identity)—and attributed to one of four protocol layers (production default; production mechanism under assembled validation; mechanism-isolation layer on reference models; experimental-layer harness). Table 1 merges the two scales. General-capability claims ("L4") are explicitly not made. Where the word "causal" is used, it means rule-like regularities inferred from temporally ordered action–outcome observations with evidence weights—not interventional identification; no do-style test is performed anywhere. *Statistical conventions.* Every comparison family declares its multiplicity correction in its own section: the beacon uses exact McNemar with no multiplicity; the mechanism campaign uses a one-sided normal-approximation Wilcoxon (rank-based, analytic tail) with Holm correction over its four confirmatory conditions (direction preregistered as FAS>B2; the

Table 1. Evidence levels × protocol layers (merged scale; used to attribute every claim in this paper).

	Level of the claim	Layer the evidence comes from
L1 representational	learned structure exists and re-activates	production default path (zero-LLM subject behavior)
L2 mechanism	measurable activation/ledger dynamics	production mechanism under experimental assembly; mechanism-isolation layer (reference models distilled from production semantics)
L3 behavioral	stored knowledge changes action selection	assembled consumer channels; decision-head harness (experimental layer, never attached to production)
L4 general capability	—	*explicitly not claimed anywhere in this paper*

frozen implementation defines the one-sided p as half the two-sided p regardless of sign, so with the all-negative ΔJCG observed here the reported Holm value quantifies the reverse-direction tail, and the preregistered-direction tail is its complement, $\approx$1); the relation-direction diagnosis is a single declared family of 21 tests; the controls experiment applies Bonferroni m=8; the persistent-state experiment tests three preregistered hypotheses; campaign v2 applies Holm within each experiment. One-sided tests are used only where the test direction was fixed before that test was run: the mechanism campaign C-1–C-4 and the persistent-state H5 are preregistered, while E2/T are post-hoc repair/replication experiments whose direction was fixed at design time but which are not part of the frozen campaign protocol v1.0; all other reported tests are two-sided, and exact methods are used at $n \leq 25$. Table 2 lists every family with its test, sidedness, pairing, and correction (full version with implementation provenance: Supplementary Material S18).

3.7.3 The mechanism-campaign protocol (preregistered)

The core incremental-value campaign studies the *incremental* value of spreading activation over flat retrieval. Four synthetic subject topologies are used (3 inputs/100 nodes/25% density; 3 inputs/300 nodes/50%; 5 inputs/300 nodes/50%; 5 inputs/1000 nodes/75%). Four candidate generators are compared: **B1** direct injection (graph-free); **B2** *flat path-product*—a closed-form approximation of retrieval on the same structure (per input–candidate pair, the best simple-path edge-weight product, summed over inputs; same structural information, no iterative propagation; note this is a closed-form surrogate, not an embedding retriever—the beacon task's flat control in Section 4.1 is a genuine sentence-embedding retriever); **B3** random activation (equal-energy control); **B4** oracle (ground truth, used only for generator-correctness prechecks; MRR=1.0 on all four topologies). The primary metric JCG (joint convergence gain) measures multi-input convergence onto shared targets. The preregistered confirmatory conditions C-1–C-4 (n=50 paired seeds each) use one-sided paired Wilcoxon with Holm correction to test H3 (*diffusion's JCG exceeds flat's*). Ancillary families (all run and archived): insertion-order permutations, label/ID permutation (hidden oracle), a no-joint-target counterfactual, a negative-edge test, a weight sweep $w \in \{0.1..0.9\}$, noise/distractor sweeps, and combinational-topology detectors. Condition-level numbers are in the Supplementary Material (Table S6); the preregistration ships with the artifacts.

3.7.4 LLM decision-head protocol (routing experiments)

One family places a real LLM at the decision point (single model, temperature 0, one call per decision). Menu, environment state, task facts, experience stream, and per-decision budgets are identical across conditions; the only difference is the context the model sees: (A) none; (B) raw last-8 experiences; (C) embedding-similarity retrieval top-5; (D) FAS mechanism-selected context (graph entry, propagation, Top-k focus, plus cognitive demand/gap/routing annotations); (E) random memory entries (counterfactual control). Four task classes are used, all fully observable at decision time. The repaired official campaign runs on the production edge-building path with nine fail-fast harness invariants (520/520 runs valid). The campaign is positioned as exploratory: it measures context-selection value, and the FAS loop does not control execution.

3.7.5 Reproducibility

Fixed seeds; single-process deterministic execution; no wall-clock-dependent logic in the measurement path (one documented exception above); archived raw runs and content summaries; all regression checks

Table 2. Statistical conventions, one row per comparison family (full version with implementation locations in Supplementary Material S18). "Status": pre = preregistered; post = post-hoc; desc = descriptive (no test). All p values in this paper are produced by the test named in this table.

Family	Test	Sidedness	paired n	Correction (m)	Status
Beacon success (§5.1)	exact McNemar	two-sided	20	none	pre
Beacon decisions (§5.1)	paired Wilcoxon	two-sided	20	none	pre
Mechanism ΔJCG, C-1–C-4 (§5.3)	Wilcoxon, normal approx. (analytic tail)	one-sided; reported Holm value is the *reverse*-direction tail (preregistered-direction tail ≈1)	50	Holm (4)	pre
Mechanism secondary (MRR, JCG>0) (§5.3)	paired Wilcoxon	two-sided	50	none (secondary family)	pre
Shared-hub sweep (§5.3)	exact McNemar; Wilcoxon	two-sided	30	none (frozen design)	pre
Relation-direction diagnosis (§5.3)	Wilcoxon (exact)	two-sided	10	single declared family (21 tests)	pre
Campaign v2 C/I/B (§5.5)	McNemar; Wilcoxon; descriptive counts	two-sided; B descriptive	10	Holm (within experiment)	pre
Campaign v2 E2/T (§5.5)	Wilcoxon (exact)	one-sided (direction fixed at design time; not in protocol v1.0)	10 / 9	Holm (within experiment)	post
Controls C3/C4 (§5.6)	exact McNemar; Wilcoxon	two-sided	10	Bonferroni (8)	pre
Persistent state H1/H3 (§5.6)	exact McNemar	two-sided	10	none (three hypotheses)	pre
Persistent state H5 (§5.6)	Wilcoxon	one-sided	10	none	pre
Routing (§5.7)	paired Wilcoxon	two-sided	20	Bonferroni (280)	post
Cross-episode recombination (§5.7)	McNemar; Wilcoxon	two-sided; one-sided (greater)	30	within experiment	pre
Determinism re-run (§5.4)	agreement counts	—	20	—	desc
Presentation-equivalence revision (§5.8)	none (stopped at gate; 0 runs)	—	0	—	post

pass at the time of writing: 82 unit tests in the pytest suite and 108/108 checks in the regression-script runner. A determinism re-run check for the LLM decision-head family is reported in Section 4.4.

4 RESULTS

4.1 Persistent relational reuse: the information-asymmetric beacon task

Task. The beacon task operationalizes information asymmetry: the knowledge needed to solve the task exists *only* in experience-written memory, in no input prompt. The target recipe is stripped from all task facts ("closed book"); the crafting menu lists `craft_X` only while item X's context is present (menu gating); goal entities are excluded from activation seeding. Four conditions, n=20 paired seeds, temperature 0, all 80 runs valid: **fas_full** (persistent graph + activation context), **D-noact** (same graph context with activation dynamics disabled), **D-flat** (flat retrieval without activation aggregation), **llm_direct** (no memory context).

Results. fas_full completes 20/20; llm_direct completes 0/20 (McNemar exact $p=2\times10^{-6}$; every decision window is used up without unlocking the chain). **D-noact**—structure present, runtime access absent—fails 0/20: structure alone is not usable access. **D-flat**—activation-free flat aggregation—fully reproduces the benefit (20/20; decisions mean 2.1 vs 2.15, no statistical difference, Wilcoxon p=0.72). First-action identity gives a mechanism fingerprint: fas and D-noact both choose `gather_oak_log` 20/20, direct only 17/20—the difference is not in first-action choice but in subsequent unlocking.

Two readings must be kept separate. **(a) Persistent relational reuse produces behavioral benefit** on a high-noise, recipe-opaque, information-asymmetric task. The 0/20 baselines are, by task design, an *existence verification*: closed-book gating makes the recipe unavailable to context-free conditions by construction, so the fas-vs-direct contrast demonstrates that the pipeline delivers the required knowledge through memory, not that memory context is "statistically significantly better" in any population sense. **(b) The necessity of activation is not supported**: D-flat's complete reproduction joins the mechanism

Table 3. Beacon task results (n=20 paired seeds per condition; 80 runs). All numbers from the archived campaign analysis.

Condition	Completed	Decisions (mean, median)	First action `gather_oak_log`	Contrast
fas_full	20/20	2.15, 2	20/20	vs direct: McNemar $p=2\times10^{-6}$
D-flat (embedding retrieval)	20/20	2.1, 2	19/20	vs fas: p=1.0; decisions Wilcoxon p=0.72
D-noact	0/20	8.0, 8	20/20	vs fas: McNemar $p=2\times10^{-6}$
llm_direct	0/20	8.0, 8	17/20	—

campaign of Section 4.3—flat retrieval suffices on reachable tasks.

What kind of retriever is D-flat? D-flat is a genuine sentence-embedding retriever: it encodes all graph node identifiers with a public, locally runnable sentence-transformer model (BAAI/bge-small-zh-v1.5) and ranks them by cosine similarity to the task text—no activation, no propagation, no path structure. A zero-LLM accessibility check (Supplementary Material S6) verifies the enabling condition directly: for the beacon goal text, the recipe node `item:oak_planks` ranks 2nd and the resource node `item:oak_log` 4th among all 44 graph nodes, inside the Top-8 context window, for all 20 seed assemblies. A complete real case study of the wood-to-plank chain (practice experience, write-back, persistent graph, beacon behavior) is shown in Supplementary Figure S3.

4.2 Write-back isolation, persistence, and consumption

Write-back isolation. If written-back structure were a rearrangement of injected knowledge, disabling write-back would change nothing. The isolation experiment breaks graph write-back on a single switch: perception, actions, and environment are identical; two lineages practice the same four-step chain for eight rounds in the same world. With write-back off, the final graph contains **0** learned `action:/change:` nodes; with write-back on, **13** (Supplementary Figure S1a). Sorting the 15 promoted relations of the write-back-on lineage (Supplementary Table S4): 11 substantive action–effect associations ("mining yields items and removes a block"), 4 degenerate relations (action pointing at its own success event—the promotion rules do not distinguish this class; we report rather than hide it), and 2 substantive pairs polluted by the attribution window (downstream craft effects attributed to the upstream gather). The initial recipe tables know *what can craft what*; *what execution produces* can only be established by doing.

Cross-context re-activation. Episode 1 practices the chain in one context; Episode 2 changes the surface context completely (interfering stand, different layout) with a structurally similar goal. Three conditions inherit from Episode 1: full graph (G0), flat-memory control with episodic-space nodes stripped (G1), and a clean graph (G2). Across 5/5 seeds: G0 re-activates **4 of 12** marked nodes ($a \geq 0.05$); G1 **0 of 8**; G2 has none (Supplementary Figure S2). The re-activation difference is carried exactly by the episodic event structure connecting shared entities; behavioral metrics are identical across conditions—representational/access-layer differences are separated from the behavior layer.

Ledger consumption. After 8 practice rounds of a three-step chain (ledger promotes 7 relations), the agent migrates to a structurally related four-step task with the ledger inherited (Transfer) or fresh (No-Transfer). Deterministic, 5/5 seeds: Transfer completes in 13 ticks / 8 actions; No-Transfer needs 16 ticks / 9 actions; the first action flips from a curiosity-driven detour to the experienced resource, and the difference lands exactly on the three candidate actions shared by both tasks (prior $(1.0,\ \text{obs}{=}8)$ *vs* (None, 0)). Task success is 5/5 in both conditions (Supplementary Figure S4).

Event-framework consumption. A negative event (a tree that disappears on first dig) is encoded as an event-framework node whose negative-polarity slot edge suppresses entity activation. After three warm-up ticks, the suppressed curiosity path loses the first-action competition: 5/5 control first-actions at tick 9 *vs* inhibited action at tick 6 (3 ticks earlier, 1 curious action saved). With *real* failure receipts (written only when an action genuinely fails), control retries the doomed target once (n=4, first completion attempt @17), the event-framework condition never (n=3, @15)—and because that failure mode is a transient timeout, the statistical channel is fully silent on it: the graph-side event framework is the only trace of

Table 4. Preregistered boundary analysis: key mechanism-level numbers. All values are read programmatically from the archived statistics files; full data in Supplementary Material S7–S9.

Test	Numbers	Reading
JCG>0 (mechanism runs)	200/200 paired seeds; medians 0.0977/0.0977/0.2158/0.2177	convergence mechanism is real
ΔJCG (C-1–C-4, preregistered)	−0.402/−0.402/−0.784/−0.782; reverse-direction Holm $p=1.5\times10^{-9}$, r=0.615, 0/200 reversals	**H3 rejected**
MRR (FAS vs B2, smaller topologies)	0.0435/0.0426 vs 0.5/0.5	flat better on reachable tasks
Noise robustness	slope 0.0076 vs 0.0; AUC 0.0427 vs 0.04	no diffusion advantage
Shared-hub topology (single weight w=0.5)	FAS 30/30 vs B2 0/30, B1 0/30, B3 5/30	the one systematic exception
Weight sweep	both methods reach MRR=1.0 together at w=0.7	structural threshold, not a FAS advantage
Dilution	$\rho(\text{reach},\text{act})$: −0.445 (noise), −0.656 (distractor)	wider spread, flatter target
Shared-hub parameter sweep (84 cells × 30 seeds)	FAS strictly better in 11 cells (all $w\leq0.5$, $d\leq1$); flat strictly better in 9 (all $w\geq0.7$, $d\geq1$); tie in 64; O3 replication 30/30 vs 0/30	advantage region: weak edges × shallow convergence paths

the failure and the only source of the behavioral difference. Scope: this channel is assembled at the experimental layer and production default is off. It is evidence that graph-encoded experience *can* flip action selection, not evidence that production does so today.

4.3 Preregistered boundary analysis of activation-based access

The convergence mechanism genuinely runs. The mechanism campaign first asks whether multi-input convergence on the graph is a real phenomenon: JCG>0 on **200/200** paired seeds (each confirmatory condition n=50; median JCG 0.0977/0.0977/0.2158/0.2177); the hypothesis group ($\geq$2 inputs) reaches MRR@1=1.0. Counterfactual controls behave as required: with no joint target, all four methods score 0.0 (convergence is not a self-fulfilling prophecy); under label/ID permutation (hidden oracle, 30 seeds), target rank 22.5 is unchanged (no structural privilege); insertion order (10 permutations) gives CV=0.0; the negative-edge test leaves totals unchanged (0.0333=0.0333; total 17.41). An independent 663-run falsification campaign had first confirmed the convergence reference model (JCG>0 rate 0.970). At the access layer, structural diffusion is also distinguishable from noise and from lookup: on a four-condition access task, full diffusion produces activation quality 61.5 over a field of 39 lit nodes, diffusion-off 16.6 over 17, the equal-energy random control 45.8 over a larger structureless field, and a target-chain oracle 17.2 over 18 (Supplementary Figure S5).

Incremental value over flat retrieval: rejected. Against the closed-form flat path-product (B2), all four preregistered conditions show ΔJCG<0 (median differences −0.402, −0.402, −0.784, −0.782; paired n=50 per condition; the frozen implementation's one-sided p is half the two-sided p regardless of direction, so on these all-negative data the reported tail is the *reverse* direction (B2>FAS): $p=3.8\times10^{-10}$, Holm (m=4) 1.5×10^{-9}, while the preregistered-direction (FAS>B2) tail is its complement, $p\approx1$; effect r=0.615; seeds where FAS beats flat: **0/200**). MRR: FAS below B2 (0.0435/0.0426 vs 0.5/0.5 on the two smaller topologies; tie at 1.0 on the two larger). Noise robustness is unsupported: robustness-sweep slope 0.0076 (B2 0.0), AUC 0.0427 vs 0.04. **H3 is rejected**: under these topologies and information budgets, flat retrieval is no worse than diffusion, and usually better (Table 4, Fig. 5a).

The shared-hub exception, parameterized. The only systematic positive difference in the preregistered campaign appears in the combinational topology: multiple inputs' relevant paths converge through a *shared hub* (FAS 30/30; B2 0/30, B1 0/30, B3 5/30; single weight configuration w=0.5). Because this exception carried the paper's only positive evidence for activation, we ran a preregistered-style parameter sweep (design frozen before execution; Supplementary Material S8): joint grid over hub-path edge weight $w\in\{0.3,\ldots,0.9\}$, number of converging inputs $k\in\{2,3,4,6\}$, and private-path depth $d\in\{0,1,2\}$, 30 paired seeds per cell, FAS vs the same closed-form flat baseline, with the original single-hub configuration as a hard preflight (replicated: FAS 30/30 vs B2 0/30). The result is a mapped surface, not a point (Fig. 4):

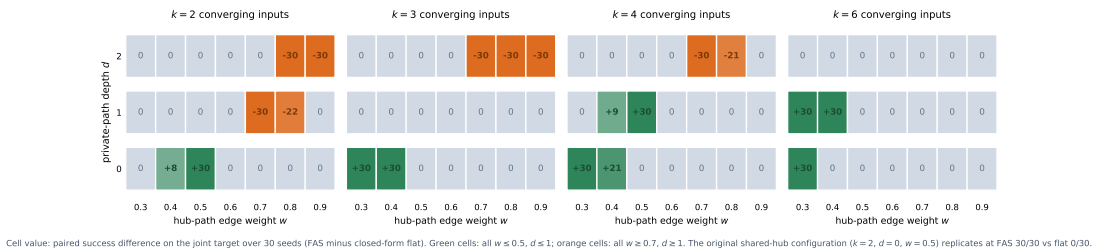


Figure 4. Shared-hub parameter sweep (84 cells × 30 paired seeds, deterministic, zero-LLM). Cell color: paired success difference (FAS minus flat) on the joint target; green = FAS strictly better, orange = flat strictly better, gray = tie; annotations give the cell counts. The FAS-advantage region is confined to weak edge weights and shallow convergence paths. All numbers read programmatically from the archived sweep outputs.

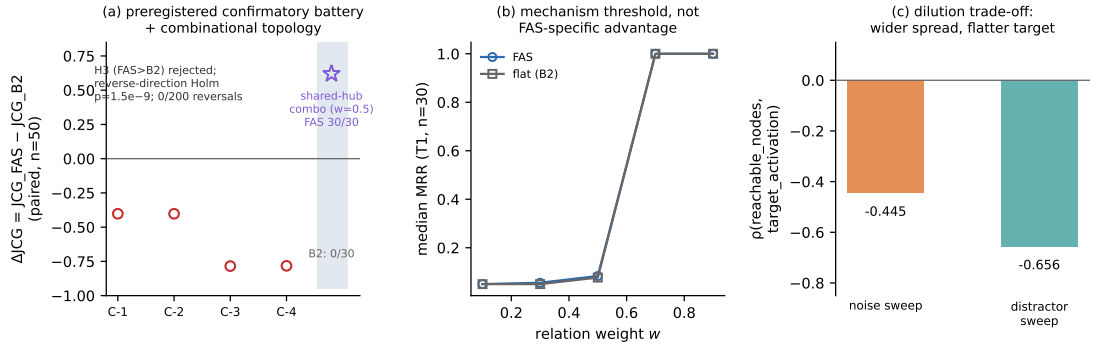


Figure 5. The diffusion applicability boundary. (a) Preregistered confirmatory family: all four ΔJCG negative (reverse-direction Holm p=1.5 × 10^{-9}); the only positive point is the shared-hub topology (star, single-weight w=0.5 design). (b) Weight sweep: both methods reach MRR=1.0 together at w=0.7. (c) Dilution trade-off: the wider the spread, the flatter the target activation (ρ=−0.445, −0.656).

FAS is strictly better in 11 of 84 cells—*all* with weak weights ($w \leq 0.5$) and shallow convergence paths ($d \leq 1$); flat is strictly better in 9 cells—*all* with strong weights ($w \geq 0.7$) and deeper paths ($d \geq 1$); the remaining 64 tie. On the O3-shaped axis (k=2, d=0), flat catches up exactly when the hub-path weight reaches 0.6 (both methods 30/30 at $w \geq 0.6$). The mechanism reading is consistent with the dilution result: convergence helps when the hub can accumulate multi-input energy faster than single paths saturate their targets, and hurts when strong edges let flat ranking resolve the target directly or when long weak paths dilute the spread.

Weight threshold, not diffusion advantage. In the campaign's weight sweep, both methods jump to MRR=1.0 together at w=0.7 (FAS 0.083@0.5→1.0@0.7; B2 0.077→1.0)—a structural resolvability threshold, not a FAS-specific advantage (Fig. 5b). The dilution trade-off completes the picture: the wider the diffusion front (more reachable nodes), the lower the target activation (ρ=−0.445 noise sweep, −0.656 distractor sweep; Fig. 5c).

Relation-direction semantics (diagnostic). A separate mechanism diagnosis (10 paired seeds; full data in Supplementary Material S9) showed that the production recipe graph is a depth-1 star forest whose relation targets are all sinks, making forward two-hop propagation structurally impossible; adding generic reverse-edge projections restores multi-hop reachability and roughly doubles the rank correlation between topological score and activation (ρ 0.325→0.645), at the cost of diluting single-node activation (0.833→0.475). All 21 paired tests give p=0.001953—the minimum attainable two-sided p at n=10 (exact Wilcoxon, all seeds concordant), which indicates uniform direction and nothing more. We report this as a design-level diagnostic (the fix is an ontology-design decision, and the production graph was not modified), not as behavioral evidence.

4.4 Determinism re-run of the decision-head family

Because the beacon and routing families use a real LLM decision head (temperature 0), we re-ran a 5-seed $\times$ 4-condition subset of the beacon campaign in a fresh output directory and compared against the archived runs. The result is an honest negative: **task outcomes** (success, per condition) match in 20/20 re-runs, but **full action sequences** match bit-for-bit in only 7/20—the remote endpoint is not bit-deterministic even at temperature 0. We therefore claim trajectory-level determinism only for the zero-LLM families (bit-identical by construction and verified by re-runs); the decision-head families are outcome-robust but not trajectory-reproducible, and no claim in this paper rests on their exact trajectories (Supplementary Material S10).

4.5 Comparative behavioral evidence (campaign v2)

The protocol above tests activation against closed-form and embedding retrieval on *mechanism* metrics. To test the same questions on *behavior*, we ran a second campaign (508 archived runs; the 390 runs from the two mid-course confound repairs are archived separately; design and full data in Supplementary Material S15–S17) in which every condition shares the same scripted experience stream, the same menu, the same LLM decision head (temperature 0), and the same decision budget—only the memory form differs. Experiment codes follow the campaign archive: C goal resumption, E goal revision, E2 event-frame wiring repair, F self-model consumer, I multi-experience convergence, B delayed credit, J long-run stability, T joint measurement; the tasks T1–T3 of the persistent-state experiment (Section 4.6) use a separate numbering (hunger recovery, resumption under context removal, failure repetition after goal invalidation).

Goal resumption under neutral instructions (C). After an open-book practice phase (obtain `oak_planks`) and an unrelated distractor phase, the test prompt gives no goal at all ("you have free choice") and the recipe facts are stripped. FAS resumes and completes the abandoned goal in **9/10** paired seeds; the full-history LLM baseline and the no-memory baseline resume in **0/10** (McNemar p=0.0039; Holm p=0.0117). This advantage is regime-dependent: with a longer experience stream that also contains a failure episode, the full-history baseline matches FAS at 10/10 (below)—the salience advantage holds in the short-history regime. Mechanism attribution: a zero-experience control reproduces the same working set (the recipe-closure prior edge `oak_log`→`oak_planks` re-lights the chain when the raw material is observed), so the effect is *observation-triggered activation over prior relational structure*—relational working-set salience—rather than an intention-lifecycle mechanism.

Goal revision is where the same mechanism costs (E). With an infeasible target (tool-gated ore), the full-history baseline stops repeating the failed gather after **1.0** failures on average and pivots immediately; FAS repeats it **10.5** times and pivots in 7/10 runs (late). A joint measurement (below) gives the direction and magnitude of this cost precisely. The two results are the same coin: state that keeps a goal salient under distraction also keeps it salient when the environment invalidates it.

Consumer repairs (E2/F). We then tested whether the failure and goal states are simply not consumed by the decision context. Wiring the production event-frame's negative-polarity encoding into the campaign assembly (failure→entity edge w=$-$0.8, parameters identical to the production config) reduced repeated failures from 10.5 to **7.2** (one-sided exact Wilcoxon p=0.0098)—statistically significant, but still far above the history baseline's 1.0 repeats; the residual gap is consistent with design-level goal pressure (constant goal text, weak ledger prior)—an attribution that remains untested. Adding a uniform self-graph goal consumer (`current_goal` $\rightarrow$ working set; production default off) turned the previously bit-identical ON/OFF behavior into a separation: with the registry cleared, the consumer-on loop re-derives the abandoned goal in 5/5 seeds and its gathering actions rise from 1.0 to **6.0** per run. The self-model's "behavioral irrelevance" was a missing-consumer artifact, not a property of the representation.

Behavioral convergence (I). On a two-chain convergence task (`oak_fence` requires `oak_planks` and `stick` learned in separate experiences), the main batch (per condition n=10) shows FAS experience-subset conditions at 0/10 (I1–I3) and 2/10 when both streams are supplied together (I4)—hence an overall 0–2/10—while the raw-history LLM baseline completes **10/10** and the retrieval baseline 8/10 (I3 *vs* I5, Holm p=0.041). In the representation batch (30 runs), context budget (k=16, 32) and a structure-enriched serialization (explicit action→outcome pairs) both remain at 0/10, locating the failure in the *serialization format*: the graph state drops the temporal narrative that the raw stream preserves (Supplementary Material S17).

Credit horizon (B) and long-run stability (J). A deterministic delay probe on the production

attribution pipeline (mechanism batch: delays 0/32/80/160 s, 10 seeds each, 40 runs) shows inventory-side action→consequence credit forming in 10/10 at every delay up to 80 s and in 0/10 at 160 s (the self-success channel is already 0/10 at 80 s)—the 90 s attribution window is the architecture's credit horizon (an intentional semantic boundary, Section 5.6). A 100-cycle stability run (zero-LLM) shows bounded graph growth (+3.7% nodes, 245→254), clamped activation, and no contamination collapse, but also zero promotion (repetition-starved episodes never reach the support threshold); short-range within-task Hebbian reinforcement of the ten prior-closure edges has been naturally reproduced in the persistent-state experiment (Section 4.6), but *long-run* (hundred-cycle, cross-episode) continuous reinforcement remains untested.

Joint measurement (T). On a single task sequence carrying both mechanisms (a failed infeasible goal followed by a neutral free phase), the event-frame encoding replicates its effect (failed-gather repetitions 12.0→7.8, one-sided exact Wilcoxon p=0.002, n=9 valid pairs), the goal consumer adds no resumption increment where the prior-edge path is already at ceiling (all conditions 10/10) and adds no inertia penalty (6.8 vs 7.8, n.s.)—the two consumers coexist, and the combined working set crowds the failed action out of the top-8 entirely. A corrected resumption probe (fresh world) shows all conditions—including the full-history baseline—at 10/10, confirming that the resumption advantage is regime-dependent and that this comparison does not isolate graph structure or activation. The behavioral trade-off curve between suppression and salience is therefore measurable, and its first two points favor suppression. (One design flaw—inventory carry-over saturating the first resumption metric—was caught and corrected with a fresh-world rerun; both datasets are archived.) E2 and T were designed after the E results and are not part of the frozen campaign protocol v1.0 (which specifies A/B/C/E/F/I/K); they are labeled post-hoc throughout, and T additionally serves as an independent replication of the wiring effect. In T's fas-ef arm, seed 0 failed twice with endpoint errors (HTTP 402 quota / connection error), was marked invalid by the runner's failure-class bookkeeping, and is excluded (n=9); the campaign protocol specifies smoke-stage harness validity with stop-on-failure but no per-run endpoint-error re-run rule—the exclusion follows the runner's pre-existing marking and is reported as such.

Together these results revise the paper's claim structure: persistent relational state demonstrably changes behavior in one direction (resumption), demonstrably costs in another (revision), is currently bottlenecked at the serialization interface (convergence), and is credit-limited by an explicit design window (delayed consequence). All negative and reverse-direction results are reported at full strength.

4.6 Persistent-state test under context removal

The campaign above varies only the memory form; the persistent-state experiment removes the *experience stream itself* from the decision context to test the behavioral role of persistence directly (preregistered; design, all conditions, and data in Supplementary Material S16). Three tasks: T1 hunger recovery (existence check), T2 goal resumption under context removal (practice a goal, then a fresh context with recipe facts stripped and only graph-derived content in the decision context), and T3 failure repetition after the environment invalidates the goal. Conditions: **FAS-full** (persistent graph context), **FAS-reset** (same stack, graph and activation reset at the phase boundary), **B0-direct** (no context), budgeted raw history **h128/h512/h2048**, **B2-rag** (embedding retrieval), and **FAS-sham** (node names without experience edges). Each hypothesis test uses n=10 paired seeds; same LLM decision head (temperature 0), same menu and decision budget.

Results. On T2, FAS-full achieves **10/10**, FAS-reset **0/10** and B0-direct **0/10** (H1/H3: exact McNemar p=0.00195—the minimum attainable at n=10 with all pairs discordant, evidencing directional consistency only, not effect size). H5 (T3): mean failure repetitions **2.8** (FAS-full) *vs* **7.7** (FAS-reset), one-sided Wilcoxon p=0.00098; B0-direct 11.2; h2048 and B2-rag both 0.0. The three preregistered hypotheses are reported with raw p values; Bonferroni over the family (m=3) leaves all three below .05 (0.0059, 0.0059, 0.0029; computed from the frozen analysis output). Note that the two failure-repetition figures in this paper come from different assemblies: campaign E's 10.5 *vs* 1.0 (Section 4.5) predates the event-frame wiring, while FAS-full here runs with the negative-polarity channel wired at the assembly layer. Mechanism observation: the ten prior-closure edges underwent Hebbian continuous reinforcement during phase 1—a short-range, within-task natural reproduction (contrasting with the 30-episode whitelisted drift of Section 4.7).

What this experiment does not show. (i) FAS-reset and B0-direct are *empty controls* (state cleared / context absent): the experiment establishes that persistence *as a whole* is causally operative, not

the independent contribution of graph structure or activation. The structure-preserving no-activation controls (flat store, activation-off) complete T2 at 10/10 each—*no difference detected* against FAS-full (n=10, limited power)—and their worse T3 behavior (12.0/11.7 repeats) is traceable to failure-evidence visibility: the control implementations structurally exclude failure information from the decision context. A frozen-preregistration revision attempting presentation equivalence was gate-stopped before any run (Section 4.8); no presentation-equivalent graph control exists yet. (ii) FAS-sham also reaches **10/10**: node-name traces plus menu gating already suffice at this task surface. (iii) The budget curve is non-diagnostic: the experience stream is only 465 tokens and observation lines embed inventory snapshots, leaking answer-bearing strings into truncated history—h128/h512/h2048 score 10/10, 10/10, 9/10 and B2-rag 10/10: whenever the experience stream stays re-readable, history and retrieval baselines match FAS on T2 (h128/h512/h2048/B2-rag: 10/10, 10/10, 9/10, 10/10 vs. FAS-full 10/10) and on T3 repeat the invalidated gather fewer times than FAS-full (h2048 and B2-rag: 0.0 vs. 2.8)—match or exceed (H2 does not hold). (iv) Single task family, n=10; all significance statements are directional claims.

4.7 Additional observations

Five further families are compressed here; full data in Supplementary Material S11–S12.

Long chain and priors. A 14-step chain completes 5/5 seeds at 315 ticks; removing graph prior edges *and* the smelting task prior gives 0/5 (the agent cannot plan out-of-recipe transitions); re-adding only the smelting prior restores the bit-identical 315 ticks; −diffusion is faster (259 ticks). With zero priors, the system never enters a learning regime: 0/25 episodes, the goal system locks onto an unreachable target, and the attribution ledger is completely untouched (0 aggregations) while the graph still grows from perception (+29 nodes)—*scaffolded grounded knowledge acquisition*: experience does not automatically become knowledge.

Hebbian, affect, intentions. Hebbian co-activation strengthening changes exactly 3 whitelisted edges (+0.074 total on the dominant edge; total-sum drift +0.079) over 30 episodes with retrieval ranks bitwise unchanged—mechanism isolation without behavioral effect. Scalar affective modulation moves activation quality monotonically with the gain (49.8 / 52.5 / 57.4 for gains 0.85/1.0/1.3; disabled is bitwise identical to neutral) but moves no action-level metric. The continuous-intention lifecycle forms, competes, and decays from real experience (3–4 intentions per run); the re-fire gate never triggers (field temperature 1.61–2.12 vs cold-field threshold 0.35), and the expression level is never reached (peak 0.46 $<$ threshold 0.78)—formation through decay is validated, expression is not.

Explicit routing (exploratory). On four fully-observable decision-head task classes (520/520 valid runs), FAS context shows one directional positive (competing-targets task: 19/20 vs 15/20 without context, uncorrected p=0.046; after correction over the campaign's 280 tests, $p\approx1$), one efficiency deficit (re-routing task: 4.15 vs 2.3/2.8 decisions, p=0.003/0.044), and structural token costs (2.1–3.7×). An earlier routing campaign was rejected by our own assembly audit (the harness never built edges and swallowed 521/521 exceptions) and is archived as invalid. Under this paper's standard, routing sits with the negative results.

Live observations. Two 25-minute live Minecraft sessions (zero LLM in the action path) are observational only: the first grows the graph by +100 nodes/+365 edges in discovery form; the second adds only +35/+95 with behavior dominated by learned goals; the live causal ledger (89 aggregations / 78 hypotheses / 29 promotions) comes entirely from real behavior. The causal bridge from write-back to behavioral improvement is *not* established in the live window—an honest negative observation.

Closure masking. Across many families, completion-layer metrics are systematically insensitive to internal differences (write-back lineages, re-activation conditions, and modulation conditions all complete identically). On tasks whose closure guarantees reachability, the planner alone suffices and activation-level knowledge falls outside the decision envelope; evaluations that report only closure-guaranteed completion rates will systematically fail to detect memory phenomena.

4.8 Audit-driven corrections and negative results

Four methodological corrections emerged during this work and are reported as first-class contributions:

(C) Misattributed resumption. The goal-resumption advantage (9/10 vs 0/10, Section 4.5) was initially labelled "persistent intention." A zero-experience control reproduces the same working set: the driver is the prior-closure edge re-lit by observation, not an intention-lifecycle mechanism. The effect is relational working-set salience, regime-dependent (short-history only).

(E) Assembly defect. The goal-revision deficit (10.5 failed repeats vs history's 1.0) was traced to a missing consumer: the production event-frame's negative-polarity channel was not wired into the campaign assembly. Wiring it reduced repeats to 7.2 (one-sided exact p=0.0098)—a partial repair; the residual gap is consistent with design-level goal pressure (attribution untested).

(F) Consumer gap. The self-model's goal state was written to the graph (telemetry 5/5/5) but had zero production readers. Adding a uniform state consumer changed autonomous behavior (gathers 1.0 $\rightarrow$ 6.0), confirming the write-only hypothesis.

(T3) Presentation inequivalence. The T3 advantage of FAS-full over flat-store and no-activation controls was traced to failure-evidence visibility: the control implementations structurally excluded failure information from the decision context. A frozen-preregistration revision (flat failure-first, 2-hop adjacency, 2-hop without polarity edges) was attempted and stopped at its own presentation-equivalence gate, before any LLM run: the flat failure-first control passed the gate, but under the naive 2-hop rule the failure-evidence node is structurally reachable (within a 51-node 2-hop set) yet ranked 43rd by discovery-edge weight (0.5 vs a slot-8 weight of 0.8) and truncated out of the 8-slot budget. Failure-evidence absence in graph controls is therefore jointly determined by distance, ranking weight, and budget—not hop distance alone—and information equivalence does not imply presentation equivalence. The preregistered stop rule took effect as designed; the independent contribution of graph structure and activation remains untested.

Negative results preserved: behavioral convergence (I) fails at all budgets; the credit horizon (B) is 90 s; cross-episode behavioral advantage over retrieval does not hold; history/RAG baselines match or exceed FAS whenever the experience stream is short enough to fit their budget (T2: equal at ceiling; T3: h2048/B2-rag 0.0 failure repeats vs. FAS-full 2.8).

5 DISCUSSION

5.1 What FAS demonstrates

Shown, within the exact scope of this protocol: (i) under mechanism-level scaffolding, embodied experience converts into relational structure absent from the initial graph and all injected sources (0 vs 13 nodes; 15 promotions); (ii) that structure persists and re-activates in changed contexts (4/12 vs 0/8 vs 0/0); (iii) two named consumers (ledger, event framework) measurably change behavior (13 vs 16 ticks; 5/5 first-action flips); (iv) under information asymmetry, persistent relational reuse yields a large behavioral difference (20/20 vs 0/20), fully reproducible by activation-free embedding retrieval; (v) the convergence mechanism runs (200/200) yet its incremental value over flat retrieval is preregistration-rejected (0/200 reversals), with the shared-hub exception delimited by the 84-cell sweep; (vi) under context removal, persistence as a whole is causally operative on behavior (FAS-full *vs* the graph-reset empty control: T2 10/10 *vs* 0/10, exact McNemar p=0.00195; T3 failure repeats 2.8 *vs* 7.7, one-sided Wilcoxon p=0.00098)—shown only against empty controls. **Not shown:** general intelligence, open-ended learning from scratch, universal superiority over LLM memory systems, causal understanding, universal diffusion benefits, robust routing value, a behavioral Hebbian benefit, or the independent contribution of graph structure and activation over structure-preserving controls (untested: no presentation-equivalent graph control has been realized; Section 4.8). Each item is accompanied by its evidence or its absence.

5.2 Why flat retrieval usually suffices—and when it does not

The negative results organize into a testable conditional statement rather than a slogan. When the task only asks "retrieve the related thing," flat retrieval does the same work at lower complexity (beacon: D-flat 20/20; campaign: ΔJCG<0). The mechanism-level reason is dilution: the wider the diffusion front, the lower the target activation ($\rho=-0.445$, -0.656), which also explains why $-$diffusion is faster on the long chain (259 vs 315 ticks). The sweep sharpens the exception into a region: *activation-based access pays when the task requires multi-relation convergence and the relevant edges are too weak for direct ranking to resolve the target* (all 11 FAS-advantage cells have $w\leq 0.5$, $d\leq 1$); with strong edges, flat ranking resolves everything; with deep weak paths, diffusion dilutes faster than the closed-form product decays (all 9 flat-advantage cells). The guarded hypothesis is: *when decision-relevant information is absent from current observation and the task structure requires multi-relation convergence, activation-based access has positive value over flat retrieval; when information is observable or the task is nearest-neighbor, that value is zero or negative.* Four evidence lines meet at this hypothesis—the beacon (information asymmetry), the shared-hub cells (convergence), the rejected campaign (nearest-neighbor), and the routing tasks (observable information)—none of them stamps it alone.

5.3 The role of scaffolding

The zero-prior result (0/25 with an untouched ledger) shows scaffolding is not a convenience but a precondition: the goal system locks onto an unreachable target before any attributable experience exists, and the learning channel starves. FAS is therefore *scaffolded grounded* rather than *learning from scratch*: the architecture must provide the mechanisms (attribution window, thresholds, mechanics tables) that make experience readable and writable. This holds for any system that must act before it can learn, and it is both a necessary transformation and a boundary of capability.

5.4 Why the unified graph matters

The three functional evidence chains (re-activation carried by episodic structure, exact ledger transfer onto shared candidates, suppression flips through polarity-slot edges) exist because experience, semantics, and action candidates *share one addressable structure*: in separated stores, each coupling would require custom bridging; here they are natural consequences of graph traversal and graph-based activation. The value of the unified graph is of the form *observable consequences*, and the negative results (closure masking, sub-threshold Hebbian drift) delimit what it cannot do: representational changes do not automatically become behavioral changes.

5.5 Design implication: direction semantics

The relation-direction diagnostic (Section 4.3; full data in Supplementary Material S9) generalizes into an ontology-design rule: *bake per-relation-class direction semantics into the ontology rather than assume propagation will find its way*—recipe-class relations forward, observation-co-occurrence relations bidirectional. On the measured recipe graph (44 nodes/53 edges), forward two-hop reachability is structurally zero because every relation target is a sink; the improvement after adding reverse projections is a design-level verification, not an emergent property.

5.6 Limitations

(1) All experiments run in a deterministic sandbox with synthetic topologies; paired seeds instantiate different random instances, not real-world variance, and all significance statements quantify directional consistency, not population effects. (2) The beacon's 0/20 baselines are existence verifications built into the task design (closed-book gating), not independent statistical findings. (3) Single environment family, single Minecraft-style domain; no third-party benchmarks; no external-environment validation (e.g., a Crafter-style equivalent) was performed. (4) Subject experiments use five deterministic seeds—descriptive statistics only; paired experiments (n=10, n=20, n=50) carry the directional claims with stated corrections. (5) Semantic write-back structure is not consumed by the planner; learned knowledge affects behavior through the ledger, event-framework, and re-activation channels only. (6) Zero-prior conditions test locking, not learning ability. (7) Not implemented: prospective memory, full perception pipeline, symbolic reasoning, RL policy nodes; several implemented mechanisms are not experimentally validated here. (8) Event-framework behavioral results come from experimental assembly of a production-default-off mechanism. (9) Scaffolding dependence is not quantified. (10) Experimental graphs are small (44–1000 nodes); scale limits all mechanism conclusions. (11) Routing tasks are fully observable at decision time and the negative results are conditional on that; token costs and directional differences are exploratory. (12) The mechanism campaign and relation-direction diagnosis are mechanism-level evidence without production behavioral tests; the shared-hub sweep, while preregistered-style and frozen before execution, is still confined to synthetic hub topologies with a closed-form flat surrogate, not an embedding retriever. (13) The beacon is a single task family (craft domain, closed book); cross-task generality of the information-asymmetry benefit is untested. (14) The decision-head families depend on a remote LLM endpoint at temperature 0; bit-level determinism is verified only by re-running (Section 4.4) (15) The attribution window (90 s) is a hard credit horizon: delayed consequences beyond it are never assigned (Section 4.5). (16) The decision-head context serialization drops the temporal narrative of the experience stream; behavioral convergence tasks therefore fail at this interface. (17) The resumption advantage over full-history baselines is regime-dependent (short-history regime only). (18) The persistent-state experiment (Section 4.6; FAS-full vs FAS-reset, 10/0) used a single task family with a 465-token experience stream; its primary McNemar p=0.00195 is the minimum attainable at n=10 and reflects directional consistency, not effect size; the budget curve is non-diagnostic because inventory snapshots in the observation lines leak answer-bearing strings into truncated history. (19) FAS-sham (nodes without experience edges) achieves 10/10 on the goal task, indicating that node-name traces plus menu gating

suffice at this task surface. (20) On goal completion (T2), no difference is detected between FAS and the flat-store/no-activation controls (10/10 each, n=10, limited power); FAS-full's T3 advantage over these controls is attributable to failure-evidence visibility (the control implementations exclude failure information from the decision context), so the independent contribution of graph structure or activation is *not* established. A frozen-preregistration revision attempt stopped at its own presentation-equivalence gate (the naive 2-hop rule still truncates the failure evidence), leaving the question untested; the gate also showed that (a) it checked only step-0 contexts over the Phase-1 baseline store, not Phase-3-evolved text, and (b) C1's context carries additional sections (edges/demand/gap/routing; 22 vs 10 lines) that the graph controls lack, so context structure was never fully aligned; answering the question requires a new, independently preregistered design.

6 CONCLUSIONS

On a unified cognitive graph, with mechanism-level scaffolding, embodied experience becomes new, typed, evidence-weighted relational knowledge: it persists, re-activates across contexts, and reaches behavior through named consumer channels in measurable, deterministic ways. On the information-asymmetric beacon task, persistent relational reuse yields a 20/20 vs 0/20 behavioral difference, fully reproduced without activation. Spreading activation is a real, measurable access mechanism whose incremental value over flat retrieval is preregistration-rejected everywhere except a now-parameterized shared-hub region: weak edges, shallow convergence paths. The context-removal experiment further locates persistence itself as a causal source of behavior (10/0 against the empty control), while the attribution of that effect to graph structure or activation over structure-preserving controls remains open—the audit marks this boundary explicitly, and the gate-stopped revision attempt leaves the question to future, independently preregistered work. The boundaries are equally clear: no-scaffolding learning regimes are never entered; closure-guaranteed tasks mask internal differences; diffusion's footprint is knowledge access, not action selection. We offer the architecture and the protocol—deterministic, zero-LLM, fully archived, with two reported assembly audits and a preregistered rejection—as a substrate for studying when experience becomes knowledge that changes behavior, and we hand over all negative results as part of the map.

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COMPETING INTERESTS

The author declares no competing interests.

AUTHOR CONTRIBUTIONS

Kailin Yan conceived and designed the study, implemented all software and experiments, performed the statistical analysis, prepared the figures and tables, verified the reproducibility of all reported results, and wrote and revised the manuscript.

DATA AND CODE AVAILABILITY

All experiments are deterministic and fully archived; the complete reproducibility package — raw run outputs, graph snapshots, statistics files, source code, preregistration records, and a one-command verifier that recomputes the paper's key numbers without an LLM — is openly available on Zenodo (DOI: 10.5281/zenodo.23196280) under a split license (data, archives, and documentation: CC BY 4.0; source code: MIT). Preregistrations are included in the package (also Supplementary Material S14–S16). LLM decision-head replications require an anonymized endpoint configuration that is not redistributed; they reproduce at the outcome level rather than bit-identical trajectories (Section 4.4).

REPORTING GUIDELINES

No established reporting checklist applies to this study: no human or animal subjects were involved, so ARRIVE/PRISMA/STROBE-type guidelines are not used. The practices actually followed were:

confirmatory analyses frozen in repository design documents before execution (the mechanism-campaign manifest, the persistent-state and controls preregistrations, and the presentation-equivalence revision with its gate-stopped preflight), deterministic zero-LLM re-runs, and full archiving of raw outputs; these frozen records are repository files shipped with the reproducibility package, not entries on a public registration platform, and they are summarized in Supplementary Material S14–S18.

USE OF GENERATIVE AI

During the preparation of this work the author used generative AI assistants for code debugging and LaTeX typesetting support. The AI tools were not used to generate the scientific ideas, experimental designs, analyses, or conclusions; all reported numbers were verified against the archived data with the verification scripts shipped in the reproducibility package. The author takes full responsibility for the content of the manuscript.

REFERENCES

Anderson, J. R., Bothell, D., Byrne, M. D., Douglass, S., Lebiere, C., and Qin, Y. (2004). An integrated theory of the mind. *Psychological Review*, 111(4):1036–1060.

Baddeley, A. (2000). The episodic buffer: A new component of working memory? *Trends in Cognitive Sciences*, 4(11):417–423.

Collins, A. M. and Loftus, E. F. (1975). A spreading-activation theory of semantic processing. *Psychological Review*, 82(6):407–428.

Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24(1):87–114.

Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mardis, A., Osinubi, J., and Others (2024). From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.

Gutiérrez, B. J., Sharpentine, Y., Tu, Y., Khramtsova, E., Zhu, X., Gweon, H., and Cohen, W. W. (2024). Hipporag: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Hart, J. and Goertzel, B. (2008). OpenCog: A software framework for integrative artificial general intelligence. In *Proceedings of the First AGI Conference (AGI-08)*.

Itti, L. and Koch, C. (2001). Computational modelling of visual attention. *Nature Reviews Neuroscience*, 2(3):194–203.

Laird, J. E. (2012). *The Soar Cognitive Architecture*. MIT Press, Cambridge, MA.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Packer, C., Fang, V., Patil, S. G., Lin, K., Wooders, S., and Gonzalez, J. E. (2023). Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*.

Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., and Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*.

Posner, M. I. (1980). Orienting of attention. *Quarterly Journal of Experimental Psychology*, 32(1):3–25.

Schacter, D. L., Addis, D. R., and Buckner, R. L. (2007). Remembering the past to imagine the future: The prospective brain. *Nature Reviews Neuroscience*, 8(9):657–661.

Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., and Yao, S. (2023). Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Tulving, E. (1983). *Elements of Episodic Memory*. Oxford University Press, Oxford.

Wang, G., Xie, Y., Jiang, Y., Mandlekar, A., Xiao, C., Zhu, Y., Fan, L., and Anandkumar, A. (2023). Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*.

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.

Zacks, J. M., Speer, N. K., Swallow, K. M., Braver, T. S., and Reynolds, J. R. (2007). Event perception: A mind-brain perspective. *Psychological Bulletin*, 133(2):273–293.